

OPENING / PENETRATION PROTECTION SYSTEM

Definition of an Opening/Penetration

Opening: Any opening in a Floor, Roof, Ceiling or Platform 2 inches or more in which materials or a person could fall through to the level below if left unprotected.

Penetration: Any opening created in a Wall, Floor, Roof, Ceiling or Platform regardless of size in which there may be a potential to damage an installation such as piping or conduit etc., due to penetrating the area beneath the surface of the wall, floor, roof, ceiling or platform.

General Requirements for Working on or in OPENINGS / PENETRATIONS

- **All Work in an Opening / Penetration** (i.e. Install of services such as Pipework, Electrical tray, cables and Ducts, drilling penetrations) requires a written and signed approval from the Area owner in the form of an Opening / Penetration Permit. Without Exception.
- **Forming an Opening / Penetration examples** (i.e. Cutting of concrete floor slab, removing of Interstitial walk on ceiling tiles or light fittings, cutting flooring metal decking, coring, drilling holes)
- **All Protections and Alterations to Openings / Penetrations** including Guard-rail systems MUST only be undertaken by an Approved and Qualified person (i.e. Carpenter / Metal Fabricator / Mechanical Fitter or Scaffolder) with written approval from the Area Owner in the form of an Opening / Penetration permit. **NO EXCEPTIONS TO THIS RULE.**
- **All Openings / Penetrations** MUST have safety signage posted stating "Opening / Penetration, Permit required" or similar language and if over 12 inches 100% Personal Fall Protection required inside.
- **For Concrete Floor slab** removal, it will require at a minimum a Method Statement and Risk Assessment and a Temporary Works approved design for any False Work, Scaffolding Crash Decks or non-standard scaffolding.
- **Guard-rail Opening / Penetration** for Openings left uncovered 12 inches or greater must have Protection Systems must be constructed to the following minimum specifications. Consisting of a top-rail an intermediate rail and posts and have a vertical height of 42 inches nominal from upper surface of top rail to floor. Vertical posts to be no more than 6 feet apart and constructed of material (i.e. Wood or Angle Iron or tubed steel) with a strength to withstand at least the minimum requirement of 200 Pounds force on the top rail in any direction and fitted with a 4inch vertical toe- board from top edge to floor and securely fastened to the floor.
- **When a Floor Opening / Penetration** is in the open position and being worked on, it must be barricaded. A spotter must be in place if the opening/penetration is greater than 12 inches, depending on the circumstances it may also require a barricade system and a spotter in place **on the floor level below** the work as well.
- **The Area Owner** must ensure they have a GA drawing of the area posted, detailing out the exact locations of any Openings / Penetrations and their current status (i.e. Barricaded with safety signage, Open for work or in the Closed position with protection cover fitted).

General Information

Area / location:		Date(s) work will be performed:	
Job description / Work to be performed (location of penetration on a GA drawing, Identify Gridlines, type material to be penetrated, and tools required to form Penetration) :			
Unique Identification number of Opening / Penetration _____ (i.e. Level 1 No 2 obtained from Area owner)			
Applicant Signature (Supervisor/Manager) _____ (I will adhere to the general safety requirements listed above)		Contact No.	Company:
Contractor HSE Signature _____		Contact No.	Company:

Opening / Penetration Checklist - For Openings / Penetrations in walls, ceilings, roofs, platforms, floors.

	Yes	No
Visually checked other side of walls, under floors, or through false ceilings for hazards (i.e. Live electrical cables, Process lines, Sprinklers, Data and Fire Alarm) and reviewed drawings if available. Note Drawing Numbers:		
Does area of Opening / Penetration need to be scanned for services (i.e. CAT, GPRS, EM scan?)		
Record the Work Permit Number: _____. Attach this opening/ penetration permit to the work permit.		
Ensure all electrical tools are inspected or tested		
Ensure the appropriate PPE worn beyond minimum PPE (i.e. Face Visor, Dust Mask, Ear Defenders, Vibration Gloves, Kevlar Gloves)		
All Power tools inspected, found to be fit for purpose and fitted with proper safety guards		
Is a crash deck / suspended scaffolding, protection scaffolding required and if so does it have a temporary works design approval and sign off by a competent person.		
Is a permanent hard barrier system with Danger Do-Not Enter signage attached required on all side of the Opening / Penetration to be formed and installed before the work commences		
If a permanent hard barrier system is not fitted, give a compelling reason why		
If unable to fit a hard barrier system, then is a double soft barrier system with a 6-foot exclusion zone in place with Danger Do-Not- Enter signage attached. For opening/ penetrations greater than 12 inches a Spotter MUST be in place when the Opening / Penetration is in the open position, under no circumstances can this type of barrier system be left unattended.		
Is the floor area directly below where the work is happening barricaded off with Danger Do-Not-Enter signage and a spotter in place for the duration of the Opening / Penetration forming work and any subsequent work in the Opening / Penetration.		
For Openings / Penetrations greater than 12 inches and when in the open position, ensure that personnel working within the barrier system are wearing Personal Fall Protection Equipment 100% of the time (i.e. Double lanyard and Body Harness) and each Individual anchorage point is capable of taking a 5,000 Pound force in any direction.		
Ensure this Opening / Penetration has been marked up as open on the GA drawing for the Floor by the Area Owner		
Area owner to ensure there is an approved and detailed Method Statement and Risk Assessment done for the task and that all contractors working on the Opening / Penetration have signed off on the Method Statement and Risk Assessment. (Including means of rescue)		
Applicant to ensure they have developed a detailed Job Safety Analysis (JSA) / Task Hazard Analysis (THA) with their work team at the actual Opening / Penetration work area and that they have all signed off on the JSA / THA		
Applicant to ensure they have spoken to other trade contractors who are working in the vicinity of the Opening / Penetration outlining what the work entails and the associated dangers.		
Applicant to ensure all workers involved in the activity are listed on the work permit.		
Check-list completed by the Applicant: _____		
Check-list reviewed by Area Owner: _____		

NOTE: ONCE THE CHECK-LIST IS COMPLETED AND SIGNED OFF BY THE APPLICANT AND AREA OWNER ONLY THEN CAN OPENING / PENETRATION WORK COMMENCE.

Reinstatement of Opening / Penetration Protection System

NOTE: Once the work in the Opening / Penetration is completed, ensure the protection cover and or guard-rail system is refitted by an Approved and Qualified Person (i.e. Carpenter / Metal Fabricator / Mechanical Fitter or Scaffolder) NO EXCEPTIONS TO THIS RULE		
All Opening / Penetration protection covers should be constructed from materials compatible with its surrounding environment (i.e. 3/4- inch thick Plywood or 1/4-inch Checkered plate) and capable of taking the weight of at a minimum two employees		
The Applicant signs off the permit to confirm their work in the Opening / Penetration as complete and the area in a good housekeeping state, they then inform the Area Owner. Applicant Signature: _____ Date: _____		
The Qualified and Approved person carrying out the Reinstatement of the Opening / Penetration Protection System must sign off the permit to acknowledge their work as safely complete: Signature _____ Date: _____		
The Area Owner MUST inspect the work area to ensure the Housekeeping is good and the Opening / Penetration cover and or guard- rail system is installed correctly and all safety signage is fitted correctly. They MUST also up-date the drawing detailing the current status of the Opening / Penetration Jacobs CM Signature: _____ Date: _____ (Area is clean and Openings / Penetrations Protections are fitted correctly)		